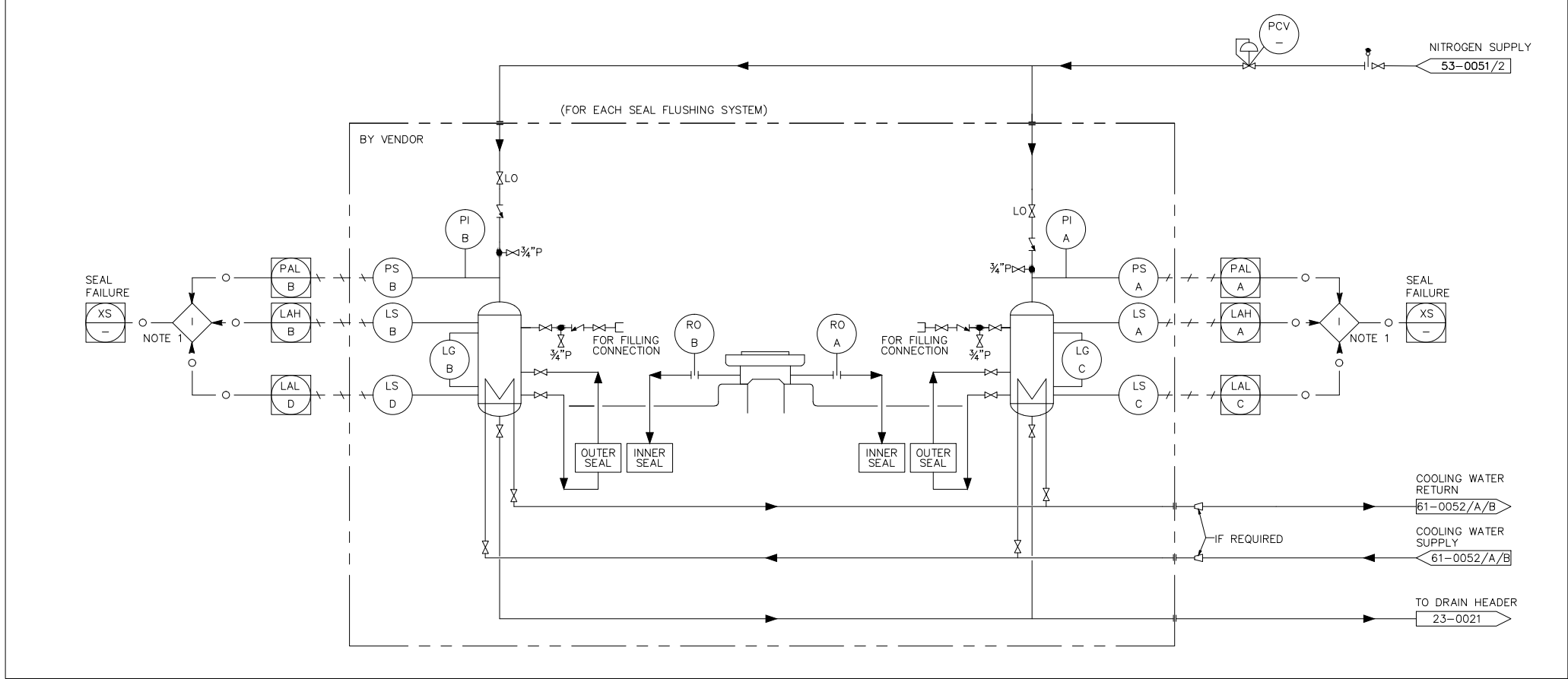


TYPICAL PUMP SEAL PLAN

API PLAN 11/53A WITH PRESSURIZED DUAL
FOR EACH SEAL FLUSHING SYSTEM



GENERAL NOTES

NOTES

1. DCS LOGIC FOR SEAL FAILURE COMMON ALARM AND THIS ALARM WILL BE ACTIVATED IN CASE OF SEAL FAILURE OF OPERATION PUMP OR SPARE PUMP.

ITEM TAG NO.	P&ID NO.	COOLING WATER SUPPLY LINE NO.	COOLING WATER RETURN LINE NO.	NITROGEN SUPPLY LINE NO.	DRAIN LINE NO.	RO-A RO-B	LG-A LG-B	LS-A LS-B	LAH-A LAH-B	LS-C LS-D	LAL-C LAL-D	PI-A PI-B	PS-A PS-B	PAL-A PAL-B		PCV	XS
P-2302A	14780-8120-25-23-0017	1 1/2"-CWS-61-052140-MS1-NI	1 1/2"-CWR-61-052141-MS1-NI	3/4"-N-53-053003-B1A1-NI	3/4"-AD-23-017017-L1A1-NI	23-1701	23-1701			23-1702	23-1702	23-1705	23-1706	23-1706		23-1702	23-1701
	14780-8120-25-23-0017	-	-	3/4"-N-53-053004-B1A1-NI		23-1702	23-1703			23-1704	23-1704	23-1707	23-1708	23-1708			23-1701
P-2302B	14780-8120-25-23-0017	1 1/2"-CWS-61-052144-MS1-NI	1 1/2"-CWR-61-052145-MS1-NI	3/4"-N-53-053005-B1A1-NI	3/4"-AD-23-017019-L1A1-NI	23-1703	23-1705			23-1706	23-1706	23-1709	23-1710	23-1710		23-1704	23-1701
	14780-8120-25-23-0017	-	-	3/4"-N-53-053006-B1A1-NI		23-1704	23-1707			23-1708	23-1708	23-1711	23-1712	23-1712			



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BASED ON

PIPING AND INSTRUMENT DIAGRAM
TYPICAL PUMP SEAL PLAN (3/4)
CONCENTRATION, DECOMPOSITION, AND
NEUTRALIZATION SECTION
UOP PHENOL PROCESS UNIT

Z1	AS BUILT	HSO	SWP	JSC	SWL	4/25/16
FS4-1	ISSUED FOR CONSTRUCTION NO HOLDS	HSO	SWP	JSC	SWL	12/29/14
S4	ISSUED FOR CONSTRUCTION NO HOLDS	HSO	SWP	JSC	SWL	10/15/14
S3	ISSUED FOR CONSTRUCTION NO HOLDS	HSO	SWP	JSC	SWL	10/8/14
S2	ISSUED FOR CONSTRUCTION NO HOLDS	HSO	SWP	JSC	SWL	9/16/14
S1	ISSUED FOR CONSTRUCTION NO HOLDS	HSO	SWP	JSC	SWL	8/20/14
R2	ISSUED FOR CONSTRUCTION WITH HOLDS	HSO	SWP	JSC	SWL	4/4/14
R1	ISSUED FOR CONSTRUCTION WITH HOLDS	HSO	SWP	JSC	SWL	2/28/14
REV.	DESCRIPTION	BY	CHK.	APPR.	PROJ. APPR.	DATE

BY	DATE	DWG.	DSGN.	CHK.	APPR.
DATE	DATE				
PROJ. NO.	DWG. NO.	REV.			
PEN	120117				Z1
PPCL	14780-8120-25-23-0001K				

SCALE